DIGITAL DESIGN AND COMPUTER ORGANISATION [BCS302]

DD MM YYYY

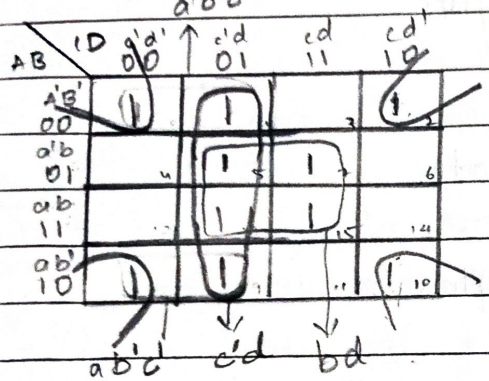
ASSIGNMENT - 01

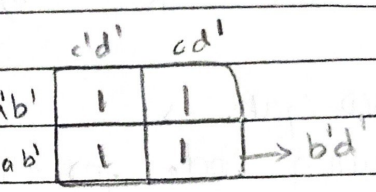
MODULE - 01

1] Minimize the following Boolean function using k-map

$F = (A, B, C, D) = \Sigma (0, 1, 2, 5, 7, 8, 9, 10, 13, 15)$

=>





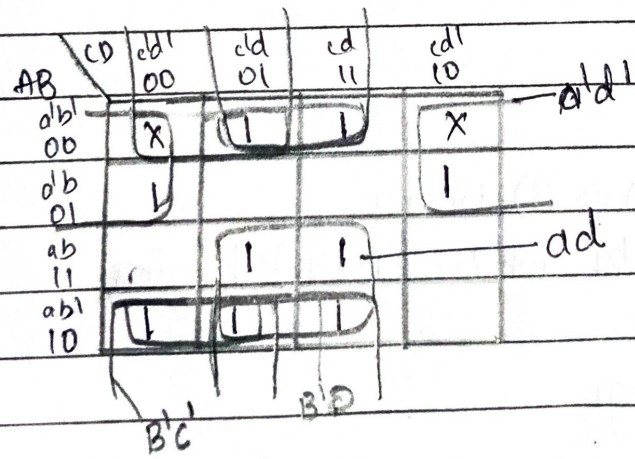
$F = ab'c' + c'd + bd + a'b'c' + bd'$

$F = c'd + bd + b'd'$

2] Minimize the following Boolean function using k-map using Don't care condition.

$F = (A, B, C, D) = \Sigma (1, 3, 4, 6, 8, 9, 11, 13, 15) + \Sigma d(0, 2, 14)$

=>



Thus, minimized boolean expression is

$F(A,B,C,D) = AD + B'D + B'C' + A'D'$

3] write verilog code for the following digital circuits.

a) AND Gate OR b) NOT gate.

=> a) AND Gate =>

Verilog code for AND Gate.

(i) Data flow Module :-

```
module AND_gate_dataflow (
output Y,
input A,
input B
);
assign Y = A & B;
end module
```

(ii) Behavioural Module :-

```
module AND_gate_behavioral (
output reg Y,
input A,
input B
);
always @(A or B) begin
if (A == 1'b1 && B == 1'b1) begin
  Y = 1'b1;
end else begin
  Y = 1'b0;
end
end module
```

(iii) Structural module :-

```
module AND_gate_structural(
output Y,
input A,
input B
);
and (Y, A, B);
end module
```

b] NOT Gate :-

Verilog code for NOT gate:

(i) data flow module :-

```
module not_gate_dataflow(
output Y,
input A
);
assign Y = ~A;
end module.
```

(ii) Behavioural Module :-

```
module. not_gate_behavioral(
output reg Y,
input A
);
always @(A) begin
Y = ~A;
end
end module.
```

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(iii) structural module :-

```
module not_gate_structural(
output Y,
input A
);
not (Y, A);
end module.
```

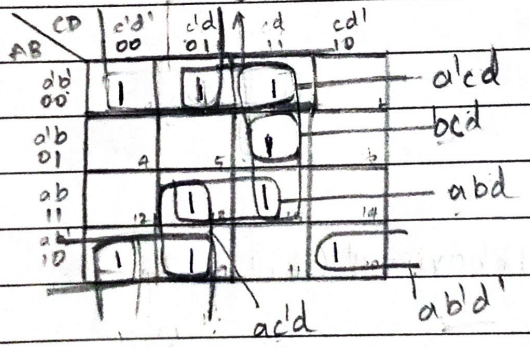
4] Simplify the following Boolean expression using k-map

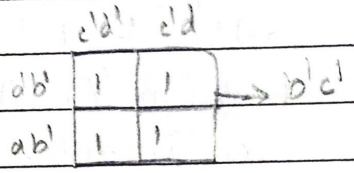
(i) $F(A,B,C,D) = \Sigma(0,1,3,7,8,9,10,13,15)$

or

(ii) $F(W,X,Y,Z) = \Sigma(0,1,2,4,5,6,7,10,15)$

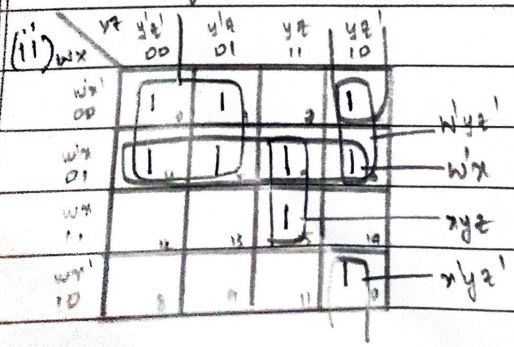
=> (i)





$F = bcd + abd + ab'd' + ac'd + b'c'a'cd + a'b'd + b'c'$

(ii)



$F = w'y' + wyz' + w'x + xyz + x'yz'$

5] Define Prime implicant and Essential Prime implicant. Give an example?

→ Prime implicant :-

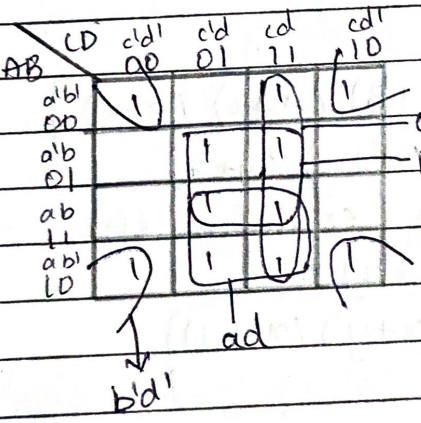
A prime implicant is a product team obtained by combining the maximum possible number of adjacent squares in the map.

Essential prime Implicant :-

An essential prime implicant is a prime implicant that covers at least once minterm that no other prime Implicant covers.

Example :-

$F(A\ B\ C\ D) = \Sigma m(0,2,3,5,7,8,9,10,11,13,15)$.



Essential prime Implicants are = BD and B'D'

prime implicants are = B'D', CD, BD, AD

6] Reduce the following Boolean expression to the minimum number of literals.

a) $X(X'+Y)$ =

$\Rightarrow XX'+XY$ ∴ $A(B'+C) = AB'+BC$. $AA'=0$.

$= 0+XY$

$= XY$

$\boxed{X(X'+Y) = XY}$

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b] $xy + x'y + yz =$

$= y(x+x') + yz$

$= y(1) + yz$

$= y + yz$

$\therefore (xy + x'y + yz = y + yz)$

$\boxed{xy + x'y + yz = y}$

$[\because x + x' = 1,\ A + AB = A]$

c] $x + x'y =$

$= x + y$

$\boxed{x + x'y = x + y}$

$[\because A + A'B = A + B]$

d] $(x+y)(x'+z)(y+z) =$

$[\because (A+B)(A'+C) = (A+B)(A'+C)(B+C)]$

$(x+y)(x'+z)(y+z) = (x+y)(x'+z)(y+z)$

since expression already contains the consensus term $(y+z)$

$\boxed{\therefore (x+y)(x'+z)(y+z) = (x+y)(x'+z)}$

e] $(x+y)(x+y') = x + yy'$

$= x + 0$

$= x$

$[\because (A+B)(A+C) = A + BC,\ AA' = 0]$

$\boxed{\therefore (x+y)(x+y') = x}$

D D M M Y Y Y Y

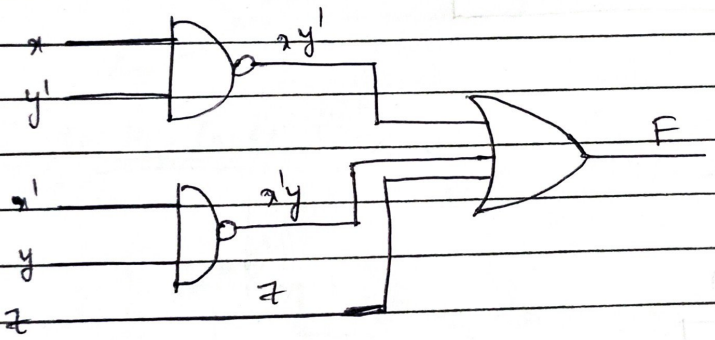
7] $F(x,y,z) = \Sigma(1,2,3,4,5,7)$ Construct the K-Map, simplify and convert into NAND gate?

⇒

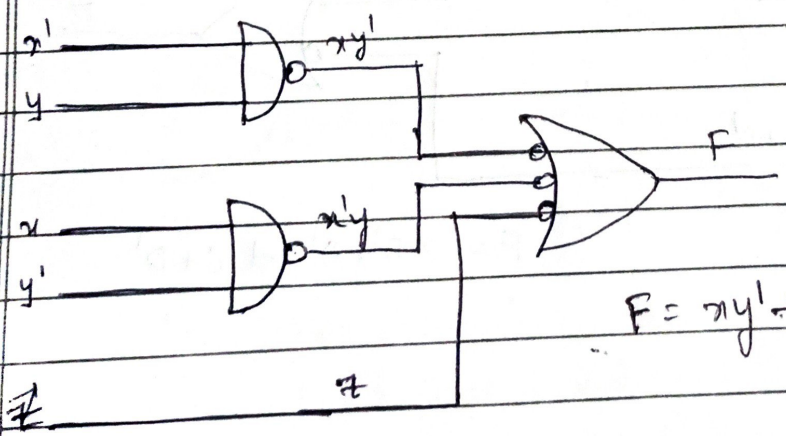
$F(x,y,z) = \Sigma(1,2,3,4,5,7)$.

x \ yz	$y'z'$ 00	$y'z$ 01	yz 11	yz' 10	
x' 0	0	1	1	1	$x'y$
x 1	1	1	1	0	
		xy'	z		

$F(x,y,z) = xy' + z + x'y$



NAND Impliments :-



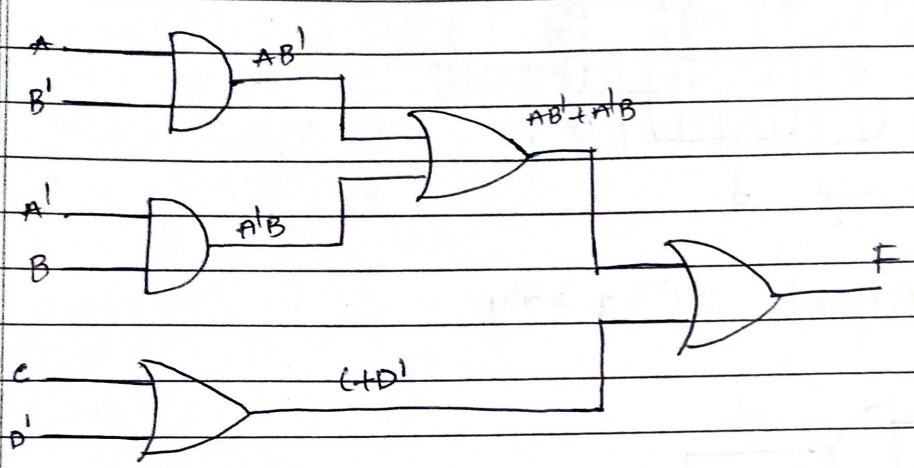
$F = xy' + x'y + z$

D D M M Y Y Y Y

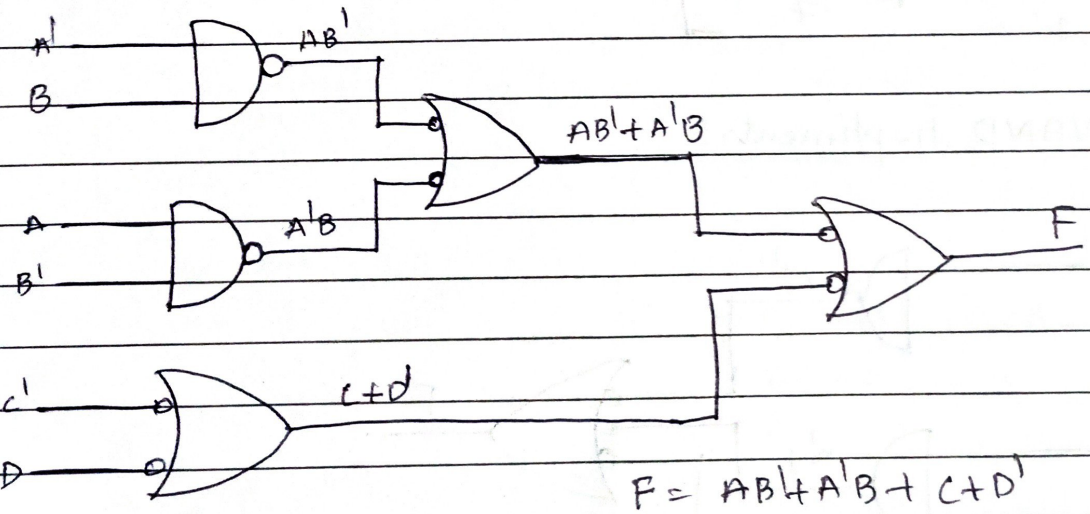
8] $F = (AB' + A'B) + (C + D')$ construct the k-map, simplify and convert into NAND gate.

→

$F = (AB' + A'B) + (C + D')$



NAND Gate:-



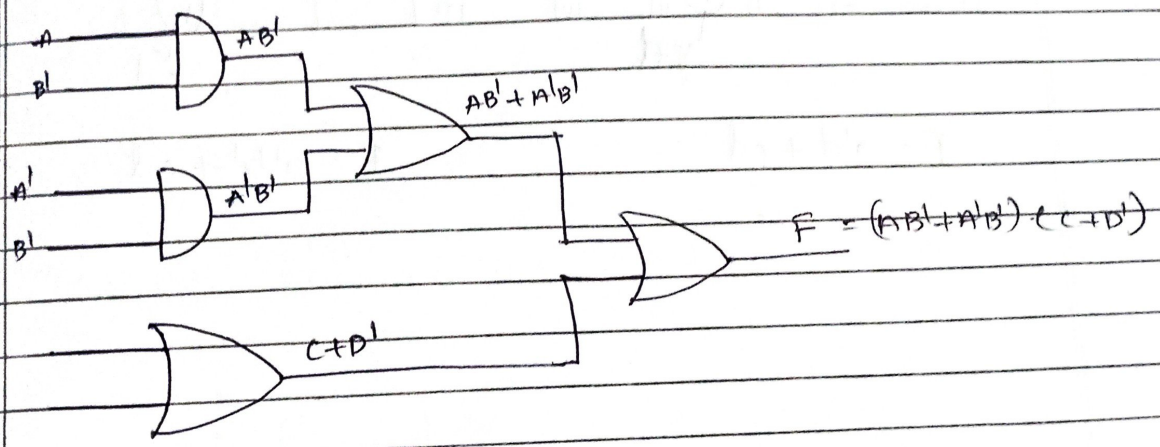
$F = AB' + A'B + C + D'$

D D M M Y Y Y Y

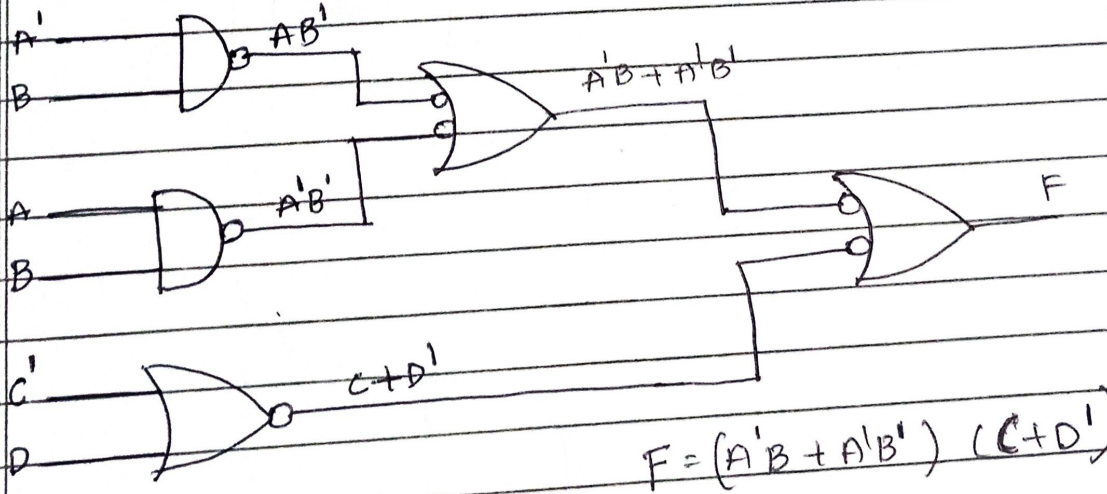
9] $F = (AB' + A'B')(C+D')$ convert into basic gates then to NOR implementation?

=>

$F = (AB' + A'B')(C+D')$



NOR Gate



$F = (A'B + A'B')(C+D')$

10] Minimize the following Boolean function using k-map using Don't care condition.

$F(A,B,C,D) = \Sigma(1,3,7,11,15) + \Sigma d(0,2,5)$

ASSIGNMENT BOOK
Laxmi
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D D M M Y Y Y Y

⇒

AB \ CD	c'd' 00	c'd 01	cd 11	cd' 10
a'b' 00	0 (0)	1 (1)	1 (3)	0 (2)
a'b 01	(4)	X (5)	1 (7)	(6)
ab 11	(12)	(13)	1 (15)	(14)
ab' 10	(8)	(9)	1 (11)	(10)

a'd

cd

$F = a'd + cd$

AB \ CD	c'd' 00	c'd 01	cd 11	cd' 10
a'b' 00	X	1	1	X
a'b 01		0	1	
ab 11			1	
ab' 10			1	

a'b'

cd

$F = a'b' + cd$